David Sebastián Rodríguez Numpaque

Astronomy Researcher — Exoplanetary Science

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About Me

BSc in Astronomy focused on exoplanetary science and computational astrophysics. Lead developer of a Bayesian, cluster-parallelized inference pipeline for exoplanetary ring detection, and co-developer of an open-source simulation package – building on a foundation of scientific software development, statistical computing, and collaborative open-source research. Additional interests include machine learning and high-energy astrophysics.

Education

BSc in Astronomy — University of Antioquia, Medellín, Colombia (2021 – *Expected graduation: June 2026*)

- Outstanding Advanced Student in Astronomy (2025)

Complementary Courses

- Research Project Design | Office of Research, University of Antioquia, 2025 – *Certified*
- The Diversity of Exoplanets | University of Geneva, 2024 – *Coursera Certified*
- Data Mining | Pontifical Catholic University of Chile, 2021 – *Coursera Certified*
- Introduction to AI | DeepLearning.AI, 2021 – *Coursera Certified*

Research Experience

SEAP (Solar, Earth and Planetary Physics) Research Group – University of Antioquia, Colombia

- **Young Researcher Intern (Oct. 2024 – Present)** — Lead developer of `PRisma`, a Bayesian pipeline that detects exoplanetary rings via *asterodensity profiling*: comparing transit-inferred stellar densities against independent estimates to reveal the "PhotoRing effect" caused by unmodeled rings.
- Built a deterministic forward model of ringed-transit observables (`exorings`) and a KDE-based likelihood, running nested-sampling inference (`dynesty`) to recover posteriors and Bayesian evidence.
- Designed and ran parallelized nested-sampling campaigns on compute-cluster resources to process TTV photometry posteriors for the Kepler-51 system (planets b and d, the archetypal "super-puff" exoplanets). Lead author of the resulting manuscript (in preparation, 2026).
- Co-developer of `Pryngles` (PlanetaRY spaNGLES), an open-source Python package for visualizing ringed exoplanets and modeling their transit light curves and polarimetric signals; contributed the modern `System` interface and its technical documentation.

Publications

1. **Numpaque, S.**, Zuluaga, J. I., Alvarado-Montes, J. A., & Kipping, D. (2026). Probing Exoplanetary Rings with Asterodensity Profiling: A PhotoRing Analysis of Kepler-51. *(In preparation)*
2. Valderrama, D. A., **Numpaque, D. S.**, et al. (2024). Artificial Intelligence in the Teaching of Natural Sciences on the Threshold of the Fifth Industrial Revolution. *Explainable AI for Education: Recent Trends and Challenges* – book chapter. `https://doi.org/10.1007/978-3-031-72410-7_9`
3. **Numpaque, S.**, et al. (2025). Contributions of Astronomy and Its Teaching to the Sustainable Development Goals: the AstrodidaXis Case. *Góndola, Enseñanza y Aprendizaje de las Ciencias* (ISSN 2665-3303). *(Submitted)*
4. **Numpaque, S.** (2024). Exploring Distant Worlds: Exoplanet Detection Workshop. *Circular Astronómica No. 999*, Colombian Astronomy Network (RAC).

Outreach & Education

Organizational Manager — AstrodidaXis Foundation, Colombia (2023 – Present)

- Coordinated outreach initiatives, research projects, and public engagement activities to strengthen astronomy education in Colombia.
- Organizing Committee member for the II & III Workshop on Astronomy Education (OAE-Colombia, AstrodidaXis, UPTC).

Mentorship Program Organizer – RECA, Colombian Astronomy Students Network (2025)

- Coordinated the national call for applications, organized panels, and managed logistics and communication for the RECA Mentorship Program.

Co-founder & Coordinator – Astronomy Research Seedbed, University of Antioquia (2025)

- Designed and led a training program for undergraduate students focused on computational astrophysics, research methods, and scientific communication.

Skills

Programming & Tools: Python (Scientific computing, OOP), C, Bash & Linux, Git/GitHub, Parallel computing, Cluster/HPC, Sphinx/RTD, Jupyter, LaTeX

Bayesian & Statistical Computing: Bayesian inference, Nested Sampling, MCMC, Forward Modeling

Machine Learning & Data Science *(in progress)*: scikit-learn

Scientific Python Packages: NumPy, SciPy, Pandas, Matplotlib, Plotly, Astropy, Astroquery

Astronomy Software: IRAF, iSpec, Rebound, Lightkurve, Batman, pyPplusS, Pryngles, Spiderman

Conferences

Speaker

- **XXI National Astronomy Meeting, RAC 2025, Colombia**
  Strengthening Research Training in Astronomy: The First Astronomy Research Seedbed at the University of Antioquia
- **VIII Colombian Congress of Astronomy and Astrophysics (CoCoA) 2024, Colombia**
  A Python Package for Modelling the Photometric Signatures of Exoplanetary Rings and Satellites
- **III International Congress on Community Environmental Education (CIEAC) 2024, Colombia**
  Exploring the Contribution of Astronomy to the Sustainable Development Goals and Community Empowerment in Boyacá

Assistant

- II Ibero-American Congress of Physics Students (CIEF), Peru — 2023
- VII Colombian Congress of Astronomy and Astrophysics (CoCoA), Colombia — 2022

Languages

- Spanish (Native) · English (B2)

References

- PhD. Jorge I. Zuluaga — jorge.zuluaga@udea.edu.co
  Full Professor, Institute of Physics, University of Antioquia
- PhD. Germán Chaparro-Molano — german.chaparro@udea.edu.co
  Assistant Professor, Institute of Physics, University of Antioquia

- PhD. Pablo A. Cuartas-Restrepo — ✉ pablo.cuartas@udea.edu.co
  Full Professor, Institute of Physics, University of Antioquia
- PhD. Oscar A. Zapata — ✉ oalberto.zapata@udea.edu.co
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